CHAPTER 17 · REFERENCE DEPTH

# Approach Control

Approach control owns the busy transition between the runway and the en-route world — sequencing arrivals down to final, launching departures up to the airways, and, where there is no ILS, talking an aircraft all the way to the runway by radar. This chapter covers IFR and VFR arrivals and departures, vectors to final, and the surveillance and precision radar approaches your original notes were missing.

---

### SYLLABUS MAP

Part III (vi) Approach control – IFR/VFR departures & arrivals, vectors to final, surveillance & precision radar approach

GHOST AVIATOR

### ◎ Learning objectives — by the end of this chapter you will be able to…

- Describe approach control's role and where it sits between Tower and Area control.
- Use IFR and VFR departure and arrival phraseology.
- Be vectored to an ILS final and read back an approach clearance.
- Fly a Surveillance Radar Approach (SRA) talkdown.

Join live classes for guided practice and doubt clearing.

- Fly a Precision Approach Radar (PAR) talkdown and apply its "do not acknowledge" rule.


**17.1 What approach control does**

**17.2 IFR & VFR departures**

**17.3 IFR & VFR arrivals**

**17.4 Vectors to an ILS final**

**17.5 Surveillance Radar Approach (SRA)**

**17.6 Precision Approach Radar (PAR)**

**☆ Numbers to memorise**

**? Question bank**





*Approach control uses surveillance radar to vector and sequence aircraft in the terminal area, bridging the gap between the aerodrome and the en-route airways.*

# 17.1 What approach control does

**IN PLAIN TERMS**

**Approach control** ("Approach", "Radar", "Director") handles aircraft in the **terminal area** — climbing departures handed up from Tower until they reach the airways, and descending arrivals handed down from Area control until they are established on final and passed to Tower. It usually has radar (a surveillance display), which is why it can **vector** aircraft and provide radar approaches.

GHOST AVIATOR

# 17.2 IFR & VFR departures

**TRANSCRIPT — IFR DEPARTURE HANDED TO APPROACH**

A/C Delhi Approach, Ghostair Alfa Bravo Charlie, passing two thousand, PAPA two departure.

ATC Ghostair Alfa Bravo Charlie, identified, climb flight level one zero zero, when ready direct PAPA.

A/C Climb flight level one zero zero, when ready direct PAPA, Ghostair Alfa Bravo Charlie.

**TRANSCRIPT — VFR DEPARTURE**

A/C Delhi Approach, VT-ABC, leaving the zone to the north, VFR, two thousand feet.

Join live classes for guided practice and doubt clearing.

ATC VT-ABC, roger, report leaving the zone, traffic is a helicopter two miles north.

# 17.3 IFR & VFR arrivals

### TRANSCRIPT — IFR ARRIVAL

A/C Delhi Approach, Ghostair Alfa Bravo Charlie, flight level one zero zero, inbound PAPA, information Delta.

ATC Ghostair Alfa Bravo Charlie, descend flight level six zero, expect ILS approach runway two seven.

A/C Descend flight level six zero, expect ILS runway two seven, Ghostair Alfa Bravo Charlie.

### "EXPECT" VS A CLEARANCE

**"Expect ILS approach runway two seven"** is advance information so you can prepare — it is **not** the approach clearance. The clearance comes later: "cleared ILS approach runway two seven". VFR arrivals are given **join instructions** as in Chapter 15.

# 17.4 Vectors to an ILS final

### WHY VECTORS

Radar lets the controller **vector** (assign headings to) arrivals so they intercept the ILS localizer at a sensible angle and spacing. The sequence is: turn onto a base heading, descend to the platform altitude, then a closing heading to intercept, then the approach clearance.

### TRANSCRIPT — VECTORS TO ILS

GHOST AVIATOR

**ATC** Ghostair Alfa Bravo Charlie, turn left heading two one zero, descend two thousand five hundred feet, QNH one zero one three.

**A/C** Left heading two one zero, descend two thousand five hundred feet, QNH one zero one three, Ghostair Alfa Bravo Charlie.

**ATC** Ghostair Alfa Bravo Charlie, closing the localizer from the left, report established. Turn left heading two five zero, cleared ILS approach runway two seven.

**A/C** Left heading two five zero, cleared ILS approach runway two seven, wilco, Ghostair Alfa Bravo Charlie.

**A/C** Ghostair Alfa Bravo Charlie, established ILS runway two seven.

**ATC** Ghostair Alfa Bravo Charlie, contact Tower one one eight decimal one.

GHOST AVIATOR

# 17.5 Surveillance Radar Approach (SRA)

## WHAT AN SRA IS

A **Surveillance Radar Approach** is a non-precision approach in which the controller uses the surveillance radar to give the pilot **azimuth (heading) guidance** and a series of **ranges with advised heights**, but **no glidepath** — the pilot maintains the correct descent by comparing the radar range with the advised height. It usually **terminates at a fixed range** (commonly 2 NM or ½ NM) from touchdown, from where the pilot continues visually.

GHOST AVIATOR

## TRANSCRIPT — SRA TALKDOWN

**ATC** VT-ABC, this will be a surveillance radar approach runway two seven, terminating at two miles. Check your minima. Turn left heading two seven zero, closing from the right.

**ATC** VT-ABC, range six miles, height should be one thousand nine hundred feet.

**ATC** VT-ABC, range four miles, height should be one thousand three hundred feet. On track.

**ATC** VT-ABC, range two miles, height should be six hundred and fifty feet, approach completed, contact Tower.

GHOST AVIATOR

## ⚠ EXAM TRAP

On an SRA the pilot is **not** normally asked to acknowledge each range/height call — the controller transmits a steady stream. The pilot flies the heading and checks the advised heights against the ranges. There is **no electronic glidepath**; the height/range advisory *is* the descent guide.

Join live classes for guided practice and doubt clearing.

# 17.6 Precision Approach Radar (PAR)

### WHAT A PAR IS

A **Precision Approach Radar** gives the pilot **continuous azimuth AND elevation guidance** — the controller watches the aircraft against both the centreline and a precise glidepath, and "talks it down" with a constant patter of corrections almost to touchdown. It is a true precision approach (used mainly by military and some civil fields).

### THE "DO NOT ACKNOWLEDGE" RULE

Because the talkdown is continuous and time-critical, the controller instructs the pilot **"do not acknowledge further transmissions"**. The pilot then simply flies the corrections and does not reply, so the channel stays clear for the steady stream of guidance.

### TRANSCRIPT — PAR TALKDOWN

**ATC** VT-ABC, this will be a precision approach runway two seven. Do not acknowledge further transmissions. Turn left heading two seven two.

**ATC** VT-ABC, approaching the glidepath … on the glidepath … on centreline.

**ATC** Slightly above the glidepath, coming down nicely … on the glidepath. On centreline.

Join live classes for guided practice and doubt clearing.

ATC Going slightly left of centreline, turn right two degrees … on centreline. On the glidepath. Approaching decision height.

### MNEMONIC — SRA VS PAR

**SRA = Steered + Range/height (no glidepath)**; **PAR = Precise (centreline + glidepath), continuous, don't acknowledge.** PAR gives *both* dimensions; SRA gives heading plus a height-check table.

GHOST AVIATOR

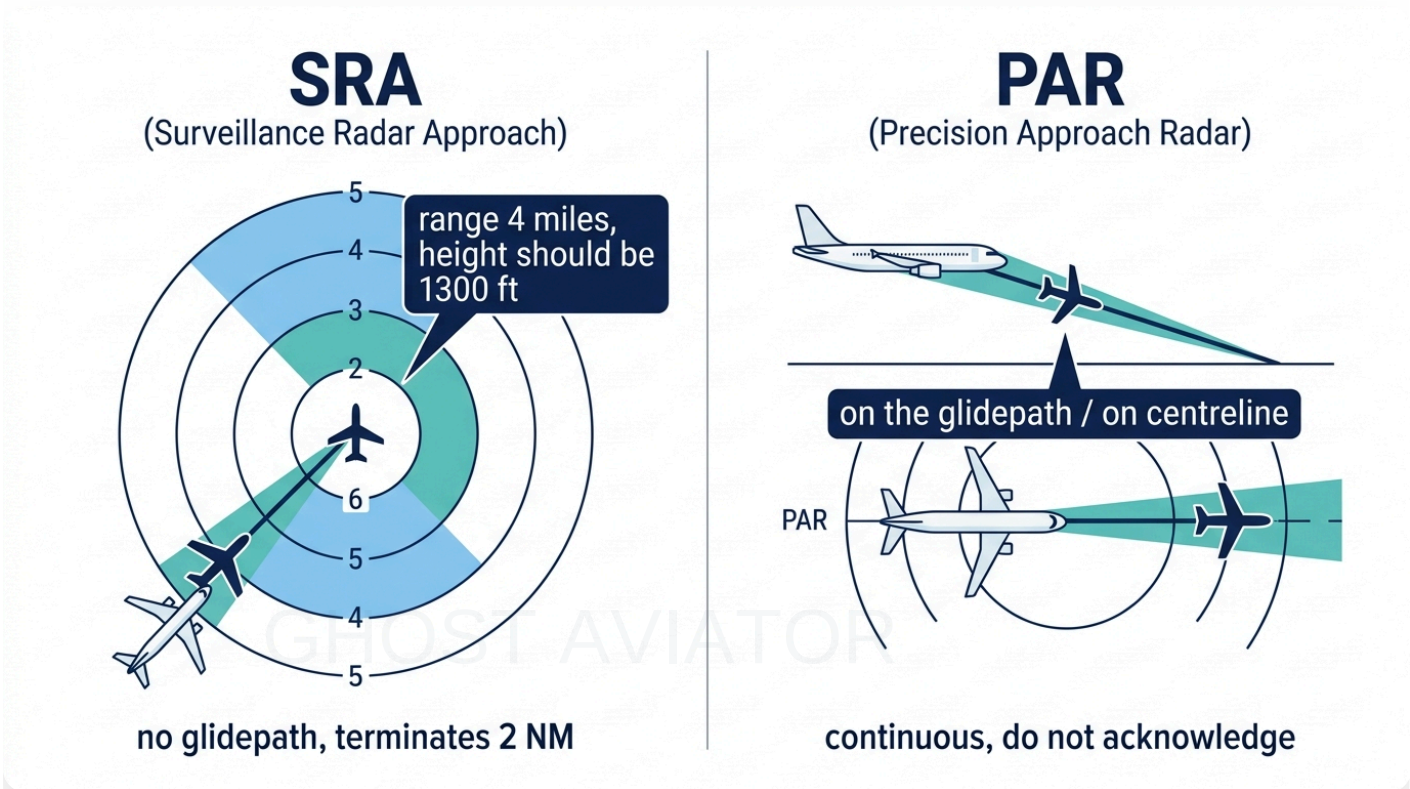



*Figure 17.1 — SRA vs PAR: what the controller can see and say. SRA provides headings and step-down altitudes, while PAR provides a continuous glidepath talkdown.*

Join live classes for guided practice and doubt clearing.

# ☆ Numbers to memorise

| Essential Facts for Chapter 17 | |
|---|---|
| **Fact** | **Value** |
| **Approach control** | Terminal area between Tower and Area control; usually radar-equipped |
| **"Expect" approach** | Advance info, NOT the approach clearance |
| **Vectors to ILS** | Base heading → descend platform → closing heading → "cleared ILS approach" |
| **SRA** | Heading + range/height advisories, NO glidepath; terminates ~2 NM (or ½ NM) |
| **PAR** | Continuous azimuth + elevation talkdown; "do not acknowledge further transmissions" |

GHOST AVIATOR

GHOST AVIATOR

# Question bank

## Part A — MCQs (click an option to check)

## 1. Approach control handles aircraft:

Only on the runway

**In the terminal area between Tower and Area control**

Only over the ocean

On the apron

GHOST AVIATOR

**Answer: In the terminal area between Tower and Area control.**

Approach sequences arrivals down to final and departures up to the airways.

GHOST AVIATOR

## 2. "Expect ILS approach runway 27" is:

The approach clearance

**Advance information, not a clearance**

A landing clearance

A vector

GHOST AVIATOR

**Answer: Advance information, not a clearance.**

"Expect" prepares you; the clearance "cleared ILS approach" comes later.

GHOST AVIATOR

### 3. To vector an aircraft, the controller assigns:

A squawk

**Headings (and often levels)**

A frequency

A call sign

GHOST AVIATOR

**Answer: Headings (and often levels).**

Vectoring is the assignment of headings using radar.

GHOST AVIATOR

## 4. A Surveillance Radar Approach provides:

Azimuth and glidepath guidance

**Heading guidance plus range/height advisories, no glidepath**

No guidance at all

Only a squawk

GHOST AVIATOR

**Answer: Heading guidance plus range/height advisories, no glidepath.**

SRA gives heading and a series of ranges with advised heights; there is no glidepath.

GHOST AVIATOR

## 5. An SRA normally terminates at:

The threshold

**A fixed range, commonly 2 NM (or ½ NM)**

10 NM

It never terminates

GHOST AVIATOR

**Answer: A fixed range, commonly 2 NM (or ½ NM).**

SRAs terminate at a set range, from where the pilot continues visually.

GHOST AVIATOR

## 6. On an SRA, the pilot maintains the correct descent by:

Following an electronic glidepath

**Comparing the radar range with the advised height**

Using GPS only

Listening to the marker beacons

GHOST AVIATOR

**Answer: Comparing the radar range with the advised height.**

The advised height for each range is the descent guide; there is no glidepath signal.

GHOST AVIATOR

## 7. A Precision Approach Radar provides:

Heading only

**Continuous azimuth and elevation (centreline + glidepath) guidance**

Range only

No talkdown

GHOST AVIATOR

**Answer: Continuous azimuth and elevation (centreline + glidepath) guidance.**

PAR gives both dimensions continuously — a true precision talkdown.

GHOST AVIATOR

## 8. On a PAR, the controller instructs the pilot to:

Acknowledge every transmission

**Not acknowledge further transmissions**

Squawk ident

Change frequency each mile

GHOST AVIATOR

**Answer: Not acknowledge further transmissions.**

"Do not acknowledge further transmissions" keeps the channel clear for continuous guidance.

GHOST AVIATOR

**9. "Closing the localizer from the left, cleared ILS approach runway 27" tells the pilot to:**

Continue past the localizer

**Intercept the localizer and fly the ILS approach**

Go around

Hold

GHOST AVIATOR

**Answer: Intercept the localizer and fly the ILS approach.**

It is the approach clearance once positioned to intercept the localizer.

GHOST AVIATOR

## 10. A VFR arrival is given:

A radar talkdown

**Join instructions (as for the circuit)**

An oceanic clearance

A SELCAL check

GHOST AVIATOR

**Answer: Join instructions (as for the circuit).**

VFR arrivals receive join instructions, then are sequenced visually.

GHOST AVIATOR

## 11. The platform/intercept altitude on vectors is given as a:

Flight level only

**Descent to an altitude with QNH, read back**

Squawk

Heading

GHOST AVIATOR

**Answer: Descent to an altitude with QNH, read back.**

Below the transition level the controller assigns an altitude on QNH, which is read back.

GHOST AVIATOR

## 12. After establishing on the ILS, the aircraft is usually:

Told to hold

**Transferred to Tower**

Told to squawk standby

Given an oceanic clearance

GHOST AVIATOR

**Answer: Transferred to Tower.**

Approach hands the established aircraft to Tower for landing.

GHOST AVIATOR

## 13. The key difference between SRA and PAR is that PAR also provides:

A squawk

**Glidepath (elevation) guidance**

A frequency change

Wind information

GHOST AVIATOR

**Answer: Glidepath (elevation) guidance.**

PAR adds continuous elevation/glidepath guidance that the SRA lacks.

GHOST AVIATOR

**14. On a PAR talkdown, "on the glidepath, on centreline" means the aircraft is:**

Too high and left

**Correctly positioned in both the vertical and the lateral plane**

Below the glidepath

Off the radar

GHOST AVIATOR

**Answer: Correctly positioned in both the vertical and the lateral plane.**

It confirms correct position on both the glidepath and the runway centreline.

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.

## 15. An IFR departure handed to Approach typically receives:

A landing clearance

**A climb clearance and identification**

A SELCAL check

Holding instructions only

GHOST AVIATOR

**Answer: A climb clearance and identification.**

Approach identifies the departure and issues climb instructions toward the airways.

GHOST AVIATOR

### 16. The reason a controller can vector and provide radar approaches is:

It has no radar

**Approach is usually radar-equipped (a surveillance display)**

It uses HF only

It is on the apron

GHOST AVIATOR

**Answer: Approach is usually radar-equipped (a surveillance display).**

Radar at approach control enables vectoring and SRA/PAR.

## Part B — Oral / viva (tap to reveal model answers)

▼ ? **What is approach control's role?**

**Model Answer:**

GHOST AVIATOR

To control aircraft in the terminal area — sequencing arrivals from Area control down to final and handing them to Tower, and taking departures from Tower and climbing them up to the airways — usually with radar, allowing vectoring and radar approaches.

▼ ? **What is the difference between "expect" an approach and being "cleared" for it?**

**Model Answer:**

"Expect ILS approach runway 27" is advance information to help the crew prepare; the

Join live classes for guided practice and doubt clearing.

clearance to fly the approach is "cleared ILS approach runway 27", given when positioned to intercept.

### ▼ ? Describe a Surveillance Radar Approach.

**Model Answer:**

A non-precision approach where the controller gives heading guidance and a series of ranges each with an advised height, but no glidepath. The pilot keeps the correct descent by comparing range with advised height, and the approach terminates at a fixed range (commonly 2 NM) from where the pilot continues visually.

GHOST AVIATOR

### ▼ ? Describe a Precision Approach Radar and its key procedural rule.

**Model Answer:**

A precision approach in which the controller gives continuous azimuth and elevation guidance — centreline and glidepath — talking the aircraft down almost to touchdown. The pilot is told "do not acknowledge further transmissions" so the channel stays clear for the continuous guidance.

### ▼ ? What is the difference between SRA and PAR?

GHOST AVIATOR

**Model Answer:**

SRA provides heading plus range/height advisories with no glidepath; PAR additionally provides continuous glidepath (elevation) guidance, making it a true precision approach.

## ⚡ 60-SECOND REVISION CARD

**Approach** = terminal area, between Tower and Area; radar-equipped → vectoring & radar approaches.

Vectors to ILS: base heading → descend to platform (QNH) → closing heading → **cleared ILS approach** → Tower.

**"Expect"** ≠ clearance. VFR arrivals get join instructions.

**SRA:** heading + range/height, **no glidepath**, terminates ~2 NM.

**PAR:** continuous centreline + glidepath, **"do not acknowledge further transmissions"**.



GHOST AVIATOR

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.